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Axiom 4 con:

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350 Induction

base case

↳ If something holds for 0, and whenever something holds for n, it holds for n+1, then something holds for all $n \geq 0$.

induction step

$S = \{n \in \mathbb{N}\}$ something holds for n

Claim: $0+1+2 \ldots + n = \sum_{i=0}^{n} i = \frac{n(n+1)}{2}$

Proof w/Axiom 4 — new way

Let S be the subset of n for which the claim holds. i.e., $S = \{n \in \mathbb{N} : \sum_{i=0}^{n} i = \frac{n(n+1)}{2}\}$ for every $n = \mathbb{N}$.

① base case

To see $0 \in S$, we compute $0 = \sum_{i=0}^{0} i$

& $\frac{0(0+1)}{2} = 0$.

Thus $\sum_{i=0}^{0} i = \frac{0(0+1)}{2}$ and $0 \in S$

② induction step

Suppose $n \in S$. Thus $\sum_{i=n}^{n} i = \frac{n(n+1)}{2}$. We compute

$$\sum_{i=0}^{n+1} i = \sum_{i=0}^{n} i + O(n) = \sum_{i=0}^{n} i + n + 1 = \frac{n(n+1)}{2} + n + 1$$

from induction hypothesis

$$= \frac{n(n+1)}{2} + \frac{2(n+1)}{2}$$

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Suppose $n \in S$. Thus $\sum_{i=n}^{n} i = \frac{n(n+1)}{2}$. We compute

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$$\sum_{i=0}^{n} i = \sum_{i=0}^{n} i + \sigma(n) = \sum_{i=0}^{n} i + n + 1 = \frac{n(n+1)}{2} + n + 1$$

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$$= \frac{n(n+1)}{2} + \frac{2(n+1)}{2}$$

$$= \frac{(n+2)(n+1)}{2}$$

Thus $\sigma(n) \in S$. Axiom 4 implies $S = \mathbb{N}$. $\square$

Proof:

Proceeding via induction, note $\sum_{i=0}^{0} i = \frac{0(0+1)}{2}$ base case

Further, if $\boxed{\sum_{i=0}^{n} i = \frac{n(n+1)}{2}}$ "something", then

$$\sum_{i=n}^{n+1} i = \frac{n(n+1)}{2} + n(n+1) = \frac{(n+1)(n+2)}{2}.$$ induction step

Our result follows. $\square$

Informal Exercise 2.

PROF. WORK

$S \subseteq \mathbb{N}$

Want $S \subset \mathbb{N}$ with $S = \mathbb{N}$ but S is closed under successors $\rightarrow$ if $n \in S$, then $\sigma(n) \in S$.

Axiom 4 tells us $0 \notin S$.

1 $S = \{1, 2, 3, 4, \ldots\} \longrightarrow \mathbb{N}^+$ pos. naturals

2 $S = \{n \in \mathbb{N} : n > 2\}$

3 $S = \{n \in \mathbb{N} : 1 \leq n \leq 5\}$

4 $S = \{n \in \mathbb{N} : n$ has a nontrivial factorization$\}$
$= \{n \in \mathbb{N} : n \geq 2\}$

* All have the form $\{n \in \mathbb{N} : n \geq k\}$ for some $k \in \mathbb{N}^+$.

Def. If $a, b \in \mathbb{N}$ are such that $\sigma(a) = b$ then we say <u>b has predecessor a</u> or <u>a is the predecessor of b</u>.

Theorem 1.

The natural number 0 has no predecessor in $\mathbb{N}$.

<u>Proof:</u>

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Proof:

For a contradiction, suppose that 0 has a predecessor $x \in \mathbb{N}$. By def. $\sigma(x) = 0$

However, this contradicts Axiom 2. $\lightning$
Thus 0 has no predecessor in $\mathbb{N}$.

$\square$

Theorem 2.

Every nonzero number has a predecessor in $\mathbb{N}$.

Proof:

We proceed via Axiom 4.

Let S be the set of natural numbers w/a predecessor union 0.

I.e., $S \in \{n \in \mathbb{N} : \sigma n\} \cup \{0\}$.
Clearly S contains 0 and is a subset of $\mathbb{N}$.

Now suppose $n \in S$.

WTS $n \in S \to \sigma(n) \in S$

We want to show $\sigma(n) \in S$. Thus we need

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We want to show $\theta(n) \in S$. Thus we need to show $\theta(n)$ has a predecessor. We need to show there is some $k \in \mathbb{N}$ such that $\theta(k) = \theta(n)$.

But of course n is the predecessor of $\theta(n)$. Thus $\theta(n)$ has a predecessor in $\mathbb{N}$, and $\theta(n) \in S$.

We conclude by Axiom 4 that $S = \mathbb{N}$. $\square$

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